Safe opponent adaptation in No-Limit Hold'em from public information

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Abstract

The strongest poker agents approximate an equilibrium and do not adapt, so they leave value unclaimed against imperfect players. We study how much a lightweight agent can gain by adapting to opponents using only public information: the betting it observes and the cards revealed at showdown. Our agent extends a Monte Carlo equity player with an opponent model that shrinks each observed frequency toward a typical player, and with bounded adjustments to ranges, bluffing, bet sizing and calling. With no evidence about an opponent, it plays exactly as the base agent. In heads-up No-Limit Texas Hold'em at 100 big blinds, over 200,000 to 300,000 hands per match, the adaptive agent wins more than the base agent against all six opponents tested: for example 65.5 against 41.4 big blinds per 100 hands (bb/100) against an opponent who over-folds, and 234 against 105 against one that switches style to defeat adaptation. It also beats the base agent directly by 8.4 bb/100 (95% interval $\pm$3.4). All opponents are synthetic and the agent is not an equilibrium player; we state these limits and outline the path to stronger baselines.

Keywords

Texas Hold'em, opponent modelling, safe exploitation, Monte Carlo simulation, imperfect-information games

1 Introduction

Agents such as Libratus and Pluribus beat top professionals by playing close to an equilibrium (Brown & Sandholm, 2018, 2019). An equilibrium strategy cannot be beaten in the long run, but it does not try to win the most. Against a player who folds too often or bluffs too much, it collects only what that player's mistakes happen to give away.

Human players exploit each other constantly, but exploitation carries a risk: a strategy fitted to an opponent can itself be exploited if the reading is wrong or the opponent changes. Methods that bound this risk exist (Johanson et al., 2008; Ganzfried & Sandholm, 2011, 2015), but most published agents either ignore opponents or adapt in ways that are hard to inspect.

We ask a narrower, practical question: how much can a simple, interpretable agent gain by adapting to opponents from public information alone, and what does it give up? Our contributions are:

1. An opponent model built only from what the whole table sees, including a way to label shown-down hands as bluffs, with each statistic shrunk toward a typical player.

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2. An adaptive agent that changes ranges, bluff frequency, bet size and calling thresholds within fixed caps, and plays identically to its base agent when it has no evidence.
3. An evaluation on commodity hardware against seven opponents, including one designed to punish adaptation, with every result reported with its 95% interval.
4. An open implementation in Java: rules engine, agents and evaluation harness.

2 Related work

Equilibrium agents. Counterfactual regret minimisation (CFR) learns approximate equilibria through self-play (Zinkevich et al., 2008). DeepStack added continual re-solving with learned value functions (Moravčík et al., 2017); Libratus combined an abstract blueprint with safe subgame solving (Brown & Sandholm, 2018); Pluribus extended the approach to six players (Brown & Sandholm, 2019); ReBeL unified search and reinforcement learning (Brown et al., 2020). None of these adapts to individual opponents.

Opponent modelling in poker. Early agents from the University of Alberta tracked opponents' action frequencies to adjust their play (Billings et al., 1998, 2002). Bayesian approaches model a distribution over opponent strategies and update it from observed play (Southey et al., 2005).

Safe exploitation. Restricted Nash response computes a strategy that exploits a model while bounding its own exploitability (Johanson et al., 2008). Data-biased responses and safe opponent exploitation bound the risk of deviating from equilibrium (Ganzfried & Sandholm, 2011, 2015).

Our position. We do not compute an equilibrium. We take a simple simulation-based agent and study bounded, interpretable adaptation on top of it, with an evaluation designed to show both the gain and its robustness. Placing the same layer on top of an equilibrium agent is future work (Section 8).

3 Problem setting

We study No-Limit Texas Hold'em with blinds of 50 and 100 chips and stacks of 100 big blinds, reset at the start of every hand. Most experiments are heads-up (two players); we also report three- and six-player tables.

Information boundary. The agent receives exactly the messages a human player's client receives: its own hole cards, every public action with its amount, the community cards, pot updates, and the cards of any player whose hand is turned over at showdown. It never sees folded or unshown cards, the deck, or another seat's private messages. In the game server this boundary is enforced in one routing class; in the evaluation harness a copy of that router is tested to produce identical messages over 300 random hands.

Goal. Given an opponent whose strategy is unknown and possibly changing, maximise the agent's win rate while (a) playing exactly as the base agent when there is no evidence, and (b) not losing to opponents who change style to exploit the adaptation.

4 Method

The agent has three parts: a base player that estimates its chance of winning by simulation, an opponent book that records what each player does, and an adaptive layer that turns the book into bounded changes to the base player's decisions.

4.1 Base agent: Monte Carlo equity

Before the flop, every one of the 1,326 starting hands is ranked by its win rate against a random hand, estimated once from 3,000 simulated deals per strategically distinct hand with a fixed seed. The agent opens a share of hands that widens toward the button, from 15% in the earliest seat to 45% on the button, and responds to raises by pot odds.

After the flop, it estimates its equity by dealing out the unseen cards 600 times (800 for the adaptive agent) against a range for each opponent. A range has a *width*, the share of starting hands (best first) the opponent could hold, and a *connectedness*, how likely their hand is to have hit the board, used to reject samples. Both are set from the opponent's actions in the current hand. The agent bets for value when its equity e exceeds a bar

that falls with the number of opponents n, and calls when e beats the pot odds by a margin m:

$$e_{\text{value}} = 0.2 + \frac{0.8}{n+1}, \tag{1}$$

$$\text{call if } e \geq \frac{c}{P+c} + m, \qquad m = \begin{cases} 0.04 & \text{on the river,} \\ 0.08 & \text{otherwise,} \end{cases} \tag{2}$$

where c is the amount to call and P the pot. The margin grows by 0.05 when the bet has been raised. These constants were chosen by comparing alternatives in the evaluation harness against rule-based opponents.

4.2 Opponent book

For each opponent the agent counts, across all hands it observes: hands dealt; voluntary entries into the pot; preflop raises; preflop raises faced and folded to; postflop bets faced and folded to; postflop actions and how many were bets or raises; and, when the opponent's cards are shown after they had bet postflop, how many such hands were weak. A shown hand counts as weak (a bluff) if its strength against the final board is below 0.45 on a 0 to 1 scale, roughly worse than a middle pair.

Each statistic is shrunk toward a typical player's value p_0 with a prior weight w, so a handful of hands proves little:

$$\hat{p} = \frac{w\,p_0 + k}{w + N}, \tag{3}$$

Table 1. The five adjustments and their caps.

Judgement	Change	Cap
Opponent's range	Width scaled by their observed raise (or entry) rate relative to typical; connectedness scaled down for opponents caught bluffing or betting often	Width $\times$0.4 to $\times$2.5; connectedness $\times$0.35 to $\times$1.3
Bluffing	Bluff when everyone still in folds often enough for a 0.7-pot bet to profit (Equation 4)	At most 65% of eligible spots; never into three or more players
Value bets	Lower bar and larger bets against players who rarely fold; the reverse against folders	Bar $\pm$0.08; size 0.4 to 1.1 of the pot
Calling	Smaller margin against frequent bluffers and aggressive bettors	Margin -0.07 to $+0.08$
Opening	Wider opening range when players behind often fold to raises	$\times$0.8 to $\times$1.5

where k is the number of times the event occurred in N opportunities. Preflop frequencies use $w = 12$, postflop frequencies $w = 8$ and the bluff rate $w = 4$, because showdowns are rarer. The typical player enters 30% of pots, raises 18%, folds to 60% of preflop raises and 45% of postflop bets, has an aggression of 0.35, and bluffs in 30% of shown-down betting lines.

4.3 Adaptive decisions

The adaptive agent overrides five of the base agent's judgements, each a function of the opponents' estimated habits (Table 1).

A bet of b times the pot breaks even as a bluff when all opponents fold with probability $b/(1 + b)$, about 0.41 for $b = 0.7$. With F the product of the opponents' fold-to-bet estimates, the agent bluffs a weak hand (equity below 0.35) with probability

$$\Pr(\text{bluff}) = \min\Big(0.65,\ s \cdot \big(0.15 + 2.5\,(F - \tfrac{b}{1+b} - 0.04)\big)\Big) \qquad (4)$$

when the edge in brackets is positive, where s is 0.55 on the flop, 0.8 on the turn and 1 on the river, since fewer bets can follow on later streets.

4.4 Safety properties

No evidence, no change. When every estimate equals the typical player, each override returns the base agent's value. A test confirms that the two agents make identical decisions in 400 random spots.

Bounded deviation. Every adjustment is clamped (Table 1), so a mistaken or manipulated read moves the agent only so far.

Gradual trust. The shrinkage prior means early reads stay close to typical.

These are engineering safeguards, not a formal guarantee of the kind restricted Nash response provides; Section 7 discusses the difference.

5 Experimental setup

Harness. Agents play each other on the same rules engine the game server uses, with no network or waiting. Every hand starts at 100 big blinds; seating is shuffled each hand and the button rotates, so no agent keeps a positional advantage. Hands are shared across worker threads, each running its own table with its own copy of every agent's opponent book, which persists for that table's whole share of the match. A fixed seed makes every match exactly repeatable.

Table 2. Opponents used in the evaluation.

Opponent	Behaviour	Role
Easy	Rule-based starting hands and hand strength; loose	Weak but reasonable
Solid	Same rules without the loose habits	Sound, non-adaptive
Simulation	The base agent of Section 4.1	Our own baseline
Caller	Never folds, never raises	Extreme: over-calls
Maniac	Bets or raises 65% of the time regardless of cards	Extreme: over-aggressive
Over-folder	Sound before the flop; afterwards folds to any bet without a strong hand	Extreme: over-folds
Chameleon	Alternates between over-folder and caller every 1,500 decisions	Designed to punish adaptation

Metric. Big blinds won per 100 hands (bb/100). The 95% interval is ± 1.96 standard errors of the per-hand result. Results whose intervals overlap are treated as not different.

Opponents. Table 2 lists the opponents. Four are extreme by design, so that the gain from each kind of adjustment can be seen.

Hardware. A 12-core consumer laptop. The harness plays about 100,000 hands per second with rule-based agents; simulation agents are slower because of the equity sampling.

6 Results

The adaptive agent won more than the base agent against every opponent, with no overlap between intervals in any match (Fig. 1, Table 3).

Table 3. Heads-up win rates of the base (simulation) and adaptive agents, in bb/100 with 95% intervals. Seed 5, 100 big blinds.

Opponent	Hands	Base agent	Adaptive agent
Caller	200,000	345.0 ± 5.4	1079.4 ± 15.4
Maniac	200,000	292.8 ± 11.0	562.8 ± 15.2
Chameleon	200,000	105.2 ± 2.8	233.8 ± 6.4
Over-folder	200,000	41.4 ± 1.3	65.5 ± 1.3
Solid	300,000	35.5 ± 1.7	57.1 ± 1.6
Easy	300,000	28.9 ± 2.3	42.1 ± 2.5

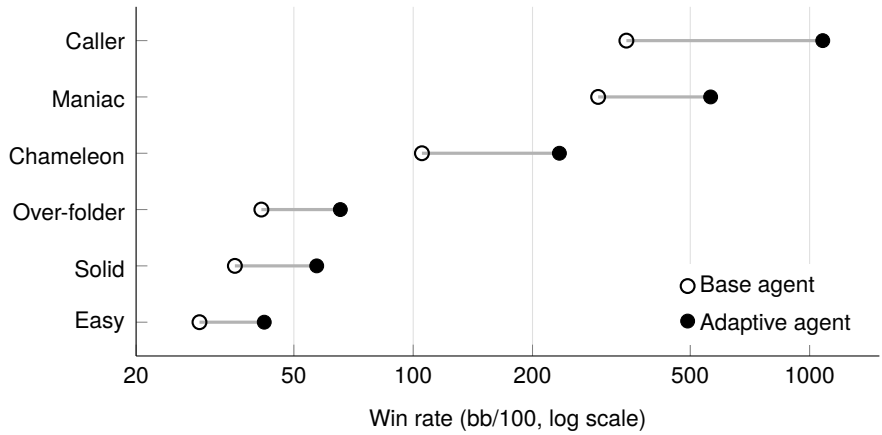


Fig. 1. Heads-up win rate of the base agent (open circles) and the adaptive agent (filled circles) against each opponent, on a log scale. The line joins the two agents' results against the same opponent. 95% intervals are smaller than the markers (Table 3).

6.1 Heads-up against each opponent

The gain is largest against the most one-sided opponents (the caller and the maniac) and smallest against the sound rule-based players, as expected. Against the chameleon, built to punish adaptation, the agent still more than doubled the base agent's win rate.

Table 4. Direct comparison and multi-player tables, in bb/100 with 95% intervals.

Match	Hands	Result
Adaptive against base, heads-up	200,000	Adaptive $+8.4 \pm 3.4$
Three-handed: the agent and two over-folders	200,000	Adaptive $+42.8 \pm 1.9$; base $+23.3 \pm 2.0$
Six-handed, sound agents only (two adaptive, two base, two Solid)	600,000	Adaptive $+7.8$ and $+3.9$ (± 2.7); base -2.1 and -3.8 (± 2.9); Solid -1.5 and -4.3 (± 2.2)
Six-handed with weak players (adaptive, base, Solid, two Easy, Caller)	600,000	Adaptive $+628 \pm 9$; base $+371 \pm 6.5$

6.2 Direct comparison and larger tables

Table 4 shows that at a table of sound agents the advantage is small and only one of the two adaptive seats clears its interval: there is little to exploit. In a live server session of 300 hands against a stand-in that only checks and calls, two adaptive agents won 165,233 and 148,557 chips against the base agent's 25,296. This is anecdotal and not used as evidence.

7 Limitations and threats to validity

Synthetic opponents. Every opponent was written by us, and four of them are extreme by design. Real players are less one-sided, so gains against people will be smaller. The chameleon tests robustness to one switching pattern, not to an adversary who models our agent.

No equilibrium baseline. The base agent is a heuristic simulation player, not an approximate equilibrium, so we cannot yet state how exploitable either agent is. The "no loss" result is against our own non-adaptive agents only.

No external benchmark or human study. We have not played public benchmark agents such as Slumbot, and the only games against people so far are informal.

Tuning on the same opponents. The base agent's constants were tuned against the rule-based opponents it is later measured against. The adaptive layer was not tuned at all.

Variance reduction. Results use plain matches, not duplicate deals or AIVAT, so intervals are wider than they need to be.

Ablations. We have not yet measured the contribution of each adjustment on its own, or how quickly the gain appears as hands are observed.

Informal safety. The caps bound how far the agent deviates but do not bound its exploitability in the formal sense of restricted Nash response.

8 Conclusion and future work

A simple opponent model built from public information, combined with bounded adjustments, roughly doubles a simulation agent's win rate against several kinds of flawed opponent, survives an opponent that switches style to defeat it, and still beats the unadapted agent directly. It needs no training, runs on a laptop, and every decision can be traced to a named statistic.

The next steps, in order, are: ablations (each adjustment switched off in turn) and learning curves (gain against hands observed); a human study with several players and a few hundred hands each; an equilibrium baseline, a self-play Monte Carlo CFR blueprint for heads-up play validated on Kuhn and Leduc poker and followed by real-time search, with the opponent model placed on top of it as a restricted Nash response so that the informal caps become a formal bound; matches against Slumbot; and reporting of compute cost (core-hours and memory) for each stage alongside published agents.

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[To be added.]

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Data availability statement

The source code of the game engine, agents and evaluation harness, and the commands and seeds needed to reproduce every match in this article, will be made publicly available at [repository link] on publication.

Responsible use

The agent was built for research and teaching on a local network. It must not be used on real-money poker sites.

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